

KnowledgeOS

Product Guide & Complete User Manual

Third-Year University Computer Science Project

Java 21 • Spring Boot 4 • PostgreSQL (pgvector & FTS) • React 19

Hybrid Retrieval-Augmented Generation Architecture

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KnowledgeOS — Complete Product Guide & User Manual

Comprehensive System Overview, High-Level Architecture, and Step-by-Step User Workflows

PART I: PRODUCT OVERVIEW & HIGH-LEVEL ARCHITECTURE

1. Product Definition & Purpose

KnowledgeOS is an authored, intelligent personal knowledge operating system engineered for students, researchers, and technical professionals. It bridges the gap between structured relational note-taking and high-precision AI retrieval by combining:

1. Deterministic Document Management: Full relational organization with collections, tags, session notes, and durable database-backed file storage.

2. Hybrid Retrieval-Augmented Generation (RAG): A dual-branch retrieval pipeline that pairs dense vector cosine similarity (`pgvector`) with exact lexical Full-Text Search (PostgreSQL FTS) using Reciprocal Rank Fusion ($k=60$).

3. Multi-Scope Grounded Synthesis: Conversational exploration across four distinct retrieval boundaries (`THIS_RESOURCE` , `SELECTED_RESOURCES` , `COLLECTION` , and `LIBRARY`) backed by verifiable source citations.

```
graph TD
    A[1. COLLECT  
Upload PDF, DOCX, TXT, Notes] --> B[2. ORGANIZE  
Tags, Collections, Smart AI Suggestions]
    B --> C[3. UNDERSTAND  
Distraction-Free Reader, Annotations]
    C --> D[4. RETRIEVE  
Hybrid RAG: Vector + Lexical FTS]
    D --> E[5. ASK  
Grounded Answers with Citations]
    E --> F[6. LEARN  
Focus Mode, Insights Dashboard]
    F --> A
```

2. The Core Knowledge Lifecycle

- 1. Collect:** Ingest academic PDFs, specifications, lecture notes, or in-app Markdown memos. Files are automatically parsed, chunked, embedded, and stored securely in PostgreSQL.
 - 2. Organize:** Categorize resources using hierarchical collections and tags, augmented by heuristic and embedding-based Smart Organization suggestions.
 - 3. Understand:** Read extracted text in a clean, high-contrast Reader interface and record inline research notes.
 - 4. Retrieve:** Query documents using conceptual language, exact alphanumeric identifiers (e.g. `CVE-2026-8819`, `RFC-9421`), or Vietnamese questions.
 - 5. Ask:** Receive syntheses from Google Gemini that are strictly grounded in your private notes, complete with verifiable citation links.
 - 6. Learn:** Reinforce concentration using timed Focus study sessions and review knowledge growth via the Insights dashboard.
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3. High-Level System Architecture

KnowledgeOS is built as a single-process **modular monolith** with clean architectural boundaries:

```

graph TB
    subgraph ClientLayer [Client Layer - Vercel]
        SPA[React 19 + TypeScript SPA]
        DesignTokens[Design Tokens & Outfit Typography]
    end

    subgraph APILayer [API Boundary - Render]
        GW[Spring Boot 4 REST API]
        SessionAuth[Session Auth & CORS Protection]
    end

    subgraph BackendModules [Modular Monolith Architecture]
        AuthMod[Auth & Profile Module]
        ResMod[Resource & Ingestion Module]
        OrgMod[Organization & Taxonomy Module]
        RAGMod[Hybrid RAG & Chat Module]
        StoreMod[Database Storage Service]
    end

    subgraph DataLayer [Persistence & AI Infrastructure]
        PG[(PostgreSQL Database  
Relational + pgvector + FTS)]
        Gemini[Google Gemini API  
Embeddings + LLM Synthesis]
    end

    SPA -->|HTTPS / JSON / Multipart| GW
    GW --> AuthMod
    GW --> ResMod
    GW --> OrgMod
    GW --> RAGMod

    ResMod --> StoreMod
    StoreMod -->|BYTEA Blobs| PG
    ResMod -->|Chunks & Vectors| PG
    RAGMod -->|Vector Cosine + FTS| PG
    RAGMod -->|Embeddings & Chat| Gemini
    AuthMod -->|User & Session Data| PG

```

Key Architectural Characteristics

- **Frontend:** React 19, TypeScript, Vite, React Router, Design Tokens (`--gs-` and `--kos-`), self-hosted `Outfit` typography, and WCAG accessibility.
- **Backend:** Java 21, Spring Boot 4, Spring Security (stateful HTTP-only session cookies), Spring Data JPA, and Bean Validation.
- **Database:** PostgreSQL with `pgvector` (768-dimensional embeddings), PostgreSQL Full-Text Search (`tsvector`/GIN), and Flyway schema migrations (V1–V13).
- **AI Infrastructure:** Google Gemini (`gemini-embedding-001` for 768-dim embeddings, `gemini-3.5-flash-lite` for grounded response synthesis).

PART II: COMPLETE STEP-BY-STEP USER MANUAL

Section 1: Getting Started & Authentication

1.1 Creating an Account

- **Goal:** Provision a new private knowledge workspace.
- **Where to Go:** `/register`
- **Steps:**
 1. Click "Sign Up" from the landing page.
 2. Enter your Full Name, Email Address, and Password (min. 8 characters).
 3. Click "Create Account".
- **Expected Result:** Account is created in PostgreSQL with BCrypt password hashing, and you are redirected to the login page.

1.2 Signing In

- **Goal:** Authenticate and establish a secure session.
- **Where to Go:** `/login`
- **Steps:**
 1. Enter your registered email and password.
 2. Click "Sign In".
- **Expected Result:** Server sets a secure `JSESSIONID` cookie and redirects you to the KnowledgeOS Home Dashboard (`/app`).

1.3 Signing Out

- **Goal:** Terminate your active session cleanly.
 - **Where to Go:** Sidebar Footer → "Sign Out" button.
 - **Steps:**
 1. Click "Sign Out".
 - **Expected Result:** Server session is invalidated and browser redirects to the login screen. Attempting to access `/app/library` directly will redirect back to `/login`.
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Section 2: Ingesting & Managing Documents

2.1 Importing a PDF Document

- **Goal:** Ingest an academic paper or technical manual.
- **Where to Go:** `/app/library` → Click "Import" button.
- **Steps:**
 1. In the upload modal, drag and drop a `.pdf` file (or click to browse).
 2. (Optional) Provide a custom Title and Description.
 3. Click "Upload & Ingest".
- **Expected Result:** The file uploads via multipart HTTP. Status transitions from `UPLOADED` → `PARSING` (Apache PDFBox) → `CHUNKING` → `EMBEDDING` (Gemini) → `READY`.
- **Notes:** Binary file bytes are stored durably in PostgreSQL `storage_blobs`.

2.2 Importing a Microsoft Word (`.docx`) File

- **Goal:** Ingest a Word document.
- **Where to Go:** `/app/library` → "Import".
- **Steps:**
 1. Select a `.docx` file and submit.
- **Expected Result:** Document text is extracted via Apache POI and indexed for search.

2.3 Importing Markdown (`.md`) or Plain Text (`.txt`) Files

- **Goal:** Ingest structured text files.
- **Where to Go:** `/app/library` → "Import".
- **Steps:**
 1. Select a `.md` or `.txt` file and submit.
- **Expected Result:** Plain text parser processes headers, lists, and code blocks into indexed chunks.

2.4 Creating an In-App Note Resource

- **Goal:** Write a new memo or meeting summary directly in KnowledgeOS.
- **Where to Go:** `/app/library` → Click "New Note".
- **Steps:**
 1. Enter a Title (e.g. "Meeting Notes on Architecture").
 2. Write your Markdown or plain text notes in the content textarea.
 3. Click "Save Note".
- **Expected Result:** Note appears immediately in the Library grid with `READY` status and is available for RAG retrieval.

2.5 Understanding Document Processing States

State Badge	Meaning	Next Transition
UPLOADED	File received and binary bytes saved in storage.	Automatically enters <code>PARSING</code> .
PARSING	Text extractor (PDFBox/POI) is reading document structure.	Enters <code>CHUNKING</code> .
CHUNKING	Text is being partitioned into 500-character segments.	Enters <code>EMBEDDING</code> .
EMBEDDING	Google Gemini is generating 768-dim vector embeddings.	Enters <code>READY</code> .
READY	Document is fully indexed for semantic, lexical, and hybrid search.	Usable in Reader and Ask.
FAILED	Ingestion encountered an unrecoverable parse or network error.	Click "Retry" to re-process.

Section 3: Workspace, Reader & Inline Notes

3.1 Opening the Resource Workspace

- **Goal:** Inspect a document's details, text, and annotations.
- **Where to Go:** Click any resource card in `/app/library`.
- **Steps:**
 1. Navigate to `/app/resource/{id}`.
- **Expected Result:** Workspace opens showing five contextual tabs: **Overview**, **Reader**, **Notes**, **Related**, and **Activity**.

3.2 Reading Full Document Text (Reader Tab)

- **Goal:** Read extracted document text without downloading original files.

- **Where to Go:** Workspace → **Reader** tab.
- **Steps:**
 1. Click "**Reader**".
- **Expected Result:** Extracted text renders with clean typography, proper line breaks, and responsive padding.

3.3 Adding & Deleting Research Notes (Notes Tab)

- **Goal:** Attach timestamped notes to a specific resource.
- **Where to Go:** Workspace → **Notes** tab.
- **Steps:**
 1. Type a note into the textarea.
 2. Click "**Add Note**".
 3. To delete, click the trash icon next to any note.
- **Expected Result:** Notes persist in the database and display chronologically.

3.4 Viewing Semantically Related Documents (Related Tab)

- **Goal:** Discover other documents in your library with similar concepts.
- **Where to Go:** Workspace → **Related** tab.
- **Steps:**
 1. Click "**Related**".
- **Expected Result:** Lists library resources ranked by vector cosine similarity. Clicking any recommendation opens that resource.

3.5 Updating Reading Progress & Favorite Status (Activity Tab)

- **Goal:** Track your reading completion and bookmark important materials.
- **Where to Go:** Workspace → **Overview** or **Activity** tab.
- **Steps:**
 1. Click the Star icon to toggle Favorite.
 2. Adjust the Reading Progress slider (0% to 100%).
- **Expected Result:** Changes save immediately via `PATCH /api/resources/{id}`.

Section 4: Organization, Tags & Collections

4.1 Creating and Assigning Tags

- **Goal:** Label resources with descriptive keywords.

- **Where to Go:** Resource Workspace → Overview tab.
- **Steps:**
 1. Type a tag name (e.g. `security`, `algorithms`) into the tag input and press Enter.
- **Expected Result:** Tag appears as a pill badge. Tags are automatically normalized to lowercase.

4.2 Grouping Documents into Collections

- **Goal:** Create folders for specific courses or research projects.
- **Where to Go:** Resource Workspace → Collection dropdown.
- **Steps:**
 1. Select an existing collection or type a new collection name (e.g. *"Computer Science 301"*).
- **Expected Result:** Resource is assigned to the collection join table.

4.3 Using Smart Organization AI Suggestions

- **Goal:** Automatically organize untagged library items.
 - **Where to Go:** `/app/library` → Click "Smart Organize" button.
 - **Steps:**
 1. Review the list of AI-generated tag and collection suggestions.
 2. Uncheck any suggestions you wish to discard.
 3. Click "Apply Suggestions".
 - **Expected Result:** Suggestions are committed in a single batch database transaction.
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Section 5: Library Search & Multi-Filter Queries

5.1 Real-Time Text Search

- **Goal:** Quickly find documents by title or description.
- **Where to Go:** `/app/library` → Search input.
- **Steps:**
 1. Type a search query (e.g. *"Architecture"*).
- **Expected Result:** Library grid updates dynamically as you type, filtering matching cards.

5.2 Filtering by Tag and Collection

- **Goal:** Narrow library view to specific topics.
- **Where to Go:** `/app/library` → Filter toolbar.
- **Steps:**

1. Select Tag dropdown → choose `oop`.
 2. Select Collection dropdown → choose "Computer Science 301".
- **Expected Result:** Only resources satisfying BOTH criteria are displayed.
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Section 6: Conversational RAG & Retrieval Scopes

6.1 Asking a Question with Hybrid RAG

- **Goal:** Query your knowledge base and receive grounded answers.
- **Where to Go:** `/app/ask`
- **Steps:**
 1. Select your desired **Retrieval Scope** (default is `LIBRARY`).
 2. Type your question (e.g. "What are the SOLID principles of OOP?").
 3. Click "Ask KnowledgeOS" (or press Ctrl+Enter).
- **Expected Result:** The system runs semantic vector search and lexical FTS, merges them via RRF, sends grounded evidence to Gemini, and renders the answer with citations.

6.2 Selecting the 4 Retrieval Scopes

KnowledgeOS provides 4 precise retrieval boundaries:

1. `THIS_RESOURCE` :
 - *Use Case:* You are studying a single paper and want answers strictly from that paper.
 - *Behavior:* Backend enforces `resource_id = :targetId`.
2. `SELECTED_RESOURCES` :
 - *Use Case:* You want to compare 2 or 3 specific documents.
 - *Behavior:* Check the desired documents in the resource picker; backend enforces `resource_id IN (:ids)`.
3. `COLLECTION` :
 - *Use Case:* You want to query all materials in a specific course folder.
 - *Behavior:* Select the collection name; backend filters by collection join.
4. `LIBRARY` :
 - *Use Case:* Global search across your entire personal knowledge base.
 - *Behavior:* Searches all `READY` documents owned by your account.

6.3 Inspecting Verifiable Source Citations

- **Goal:** Audit the factual evidence behind an AI-generated answer.

- **Where to Go:** `/app/ask` → Below the assistant response.
- **Steps:**
 1. Click any citation badge (e.g. `[1]`, `[2]`).
- **Expected Result:** A citation drawer opens showing the exact source document title, chunk index, similarity score, and verbatim extracted text. Click **"Open in Reader"** to jump directly to the document.

6.4 Continuing Multi-Turn Conversations

- **Goal:** Ask follow-up questions maintaining chat context.
- **Where to Go:** `/app/ask`
- **Steps:**
 1. Type a follow-up (e.g. *"Can you give a Java code example of the first principle?"*).
 2. Submit.
- **Expected Result:** Multi-turn history is preserved in PostgreSQL `chat_messages` and displayed in the left conversation history rail.

6.5 Handling Unsupported Questions (Anti-Hallucination)

- **Goal:** Verify behavior when documents do not contain the answer.
- **Steps:**
 1. Query an off-topic question (e.g. *"What is the recipe for chocolate cake?"*).
- **Expected Result:** The system explicitly responds: *"I cannot find information about chocolate cake recipes in your library documents"*, avoiding fabricated hallucinations.

Section 7: Focus Mode, Insights & Account Management

7.1 Using Focus Mode for Timed Study Blocks

- **Goal:** Eliminate interface distractions during deep reading.
- **Where to Go:** `/app/focus`
- **Steps:**
 1. Select a document from your library.
 2. Choose a timer interval (e.g. 25 minutes Pomodoro).
 3. Click **"Start Focus"**.
- **Expected Result:** Navigation collapses into a distraction-free stage with an active countdown timer.

7.2 Viewing Analytics in Insights Dashboard

- **Goal:** Track knowledge base growth and document composition.
- **Where to Go:** `/app/insights`
- **Expected Result:** Displays Total Resources, Total Chunks, Estimated Reading Time, and Tag breakdown.

7.3 Updating Profile and Changing Password

- **Goal:** Maintain user account credentials.
- **Where to Go:** `/app/profile`
- **Steps:**
 1. Update Display Name, enter current password and new password.
 2. Click "Update Profile".
- **Expected Result:** Profile is updated and new credentials are confirmed.

7.4 Safely Deleting a Resource

- **Goal:** Permanently remove a document and clean up all dependent records.
- **Where to Go:** Resource Workspace → Click "**Delete**".
- **Steps:**
 1. Click "**Delete**".
 2. Click the red confirmation button.
- **Expected Result:** Resource, chunks, binary storage blobs, and associated citations are safely removed without database errors.